La Consolacion University Philippines
(formerly University of Regina Carmeli)
Catmon, City of Malolos

SCHOOL YEAR 2025-2026

COLLEGE OF INFORMATION TECHNOLOGY AND ENGINEERING

I. **COURSE CODE** : **IT 101**

II. **COURSE TITLE** : **Introduction to Computing**

III. COURSE DESCRIPTION :

This course provides a foundational understanding of information technologies, with a primary focus on the architecture and functions of the electronic digital computer. Students will explore the two major components of computing systems—hardware and software—and examine the principles governing their operations. The course also covers essential topics such as computer networks, multimedia systems, information systems, the Internet, and current trends and issues in information technology. Special emphasis is placed on IT ethics, ensuring students understand the responsible use of technology in a rapidly evolving digital landscape.

Beyond technical knowledge, the course integrates Marian Augustinian values into each lesson, fostering the development of students as Christo-centric Catholic professionals. The curriculum not only aims for intellectual growth but also instills a commitment to ethical decision-making, social responsibility, and compassionate service. Students will be equipped with the knowledge and values necessary to thrive as professionals in the field of Information Technology.

IV. PRE – REQUISITE : None

V. CREDIT UNITS : 3 units

VI. COURSE LEARNING OUTCOMES :

Vision	Mission
Moved by the Spirit of the Risen Christ, La Consolacion University Philippines, envisions itself to become a catalyst for the wholistic transformational development	To achieve the above vision, LCUP commits itself to generate and make available to its Augustinian Marian students, programs and services that will make them:

of Catholic Augustinian Marian graduates rooted in Gospel values.	● *exhibit academic and technical competence in their respective areas of discipline in the local and international environment;* ● *demonstrate their knowledge, critical thinking and creativity by exploring the various avenues of learning and searching for truth;* ● *live the Gospel values in the varied aspects of their personal and professional lives;* ● *manifest their willingness to serve specially the marginalized sector of our society;* ● *demonstrate productivity in various areas of learning as responsible citizens of the society; and* *develop individuals who are resilient, inclusive and responsive to be socio-economic realities of the Philippines and the complexities of the global environment.*

EXPECTED GRADUATE ATTRIBUTES	PROGRAM OUTCOMES	COURSE LEARNING OUTCOMES
EGA 1: Critical and Creative thinker EGA 2: Competent Catholic Augustinian-Marian Professional EGA 3: Socially Responsive Steward EGA 4: Transformative Lifelong learner	PO1. Articulate and discuss the latest developments in the specific field of practice; PO2. Effectively communicate orally and in writing using both English and Filipino; PO3. Work effectively and independently in multi-disciplinary and multi-cultural teams; PO4. Act in recognition of professional, social ethical responsibility; PO5. Preserve and promote "Filipino historical and cultural heritage; PO6. Analyze complex problems, and identify and define the computing requirements needed to	CLO 1. Develop critical thinking and creativity in solving complex IT problems, leveraging a strong foundation in information and software technology, programming tools, and modern applications to create innovative solutions for managing information and network challenges. [EGA 1] [PO 1,2] CLO 2: Demonstrate ethical responsibility, humility, and professionalism in developing solutions, while maintaining integrity in distinguishing between software systems and understanding software updates. [EGA 2] [PO 4,5,6]

	design and appropriate solution; PO7. Apply computing and other knowledge domains to address real-world problems; PO8. Design and develop computing solutions using a system-level perspective; PO9. Utilize modern computing tools.	CLO 3: Address social and ethical issues in IT by ensuring responsible use of digital resources and data privacy, while identifying and implementing solutions that benefit society and serve the greater good. [EGA 3] [PO 1,2,3,4,6,7,8] CLO 4. Demonstrate the Filipino-Catholic Augustinian Spirituality through value webbing. [EGA 2] [PO 4,5]

VII. LEARNING PLAN:

SCHEDULE	INTENDED LEARNING OUTCOMES	COURSE CONTENTS	SUGGESTED LEARNING ACTIVITIES	SUGGESTED ASSESSMENT TASKS	EXPECTED VALUE/S TO BE INTEGRATED
Week 1	ILO10: Demonstrate the Augustinian, Marian and missionary Christ-centered Spirituality through value webbing. [CLO 4]	**I. COURSE ORIENTATION** A. Introduction to the Course Contents B. Course Requirements and Activities C. Course Objectives D. Grading System and Policies on Absences, Tardiness and Examinations	• Setting and sharing of Expectations from the Course • Discussion on the Vision, Mission and Goals of LCUP and of CITE • Presentation of Course Objectives and Contents, Course Requirements and Activities • Review of Policies on Absences, Tardiness, Examinations,	• Statement of Expectations • Review of Course Contents, Objectives, Activities and Assessment Tools • Review of Policies on Absences, Tardiness, Examinations, Grading System	• Interiority • Respect • Obedience

			Grading System		
Week 2-3	ILO1: identify and describe the functions of key hardware components such as the CPU, memory, storage devices, and input/output devices. [CLO 1][CLO2] ILO2: Demonstrate the ability to assemble, upgrade, and maintain a computer system, including the installation of hardware components and peripheral devices. [CLO 1][CLO2] ILO3: Understand the role of an operating system and be proficient in installing, configuring, and managing operating systems (e.g., Windows, Linux) on computer hardware. [CLO 1][CLO2]	II. **Introduction to Computer** • Hardware Concepts ➢ Computer ➢ Digital Representation ➢ Digital Electronics ➢ Digital Device ➢ Analog Device • Software Concepts ➢ Software ➢ Type of License ➢ Computer program ➢ Support module ➢ Data Module ➢ Application software ➢ System Software	• Lecture/Discussion • Work in groups to assemble a computer from individual components, such as the motherboard, CPU, RAM, storage devices, and peripherals.	**Task**: Students are given a set of hardware components (motherboard, CPU, RAM, storage, power supply, etc.) and asked to assemble a fully functional computer system. **Assessment Criteria**: Correct assembly, component placement, cable management, and system functionality after assembly.	• Appreciation • Awareness • Courage • Humility
	ILO1: Define and explain the concept of an algorithm,	**III. Algorithms and Flowcharting**	• Students will use flowcharting tools (e.g., Lucidchart,	**Assessment Criteria**: Correctness of the	• Creativity • Critical Thinking

Week 3-5	including its structure, purpose, and characteristics in solving computational problems. [CLO 1][CLO2] ILO2: Create flowcharts to visually represent the steps of an algorithm using standard flowchart symbols, demonstrating an understanding of logical processes. [CLO 1][CLO2] ILO3: Apply algorithmic thinking and flowcharting techniques to model and solve real-world problems across various domains [CLO 1][CLO2]	• Algorithm: ➢ What is an Algorithm ➢ Characteristics of an Algorithm • Flowcharting: ➢ Advantages of using flowcharts ➢ Limitations of using flowcharts ➢ Basic Symbols • Sample Flowcharts	draw.io) to visually represent the algorithm. They must use appropriate flowchart symbols (e.g., decision diamonds, process rectangles) to show the flow of logic. • Students exchange their flowcharts and algorithms with classmates for peer review. Each student reviews a peer's work for logical correctness, readability, and efficiency, offering constructive feedback. • **Task**: Provide students with an erroneous or incomplete flowchart and pseudocode. They must identify errors, correct them, and explain the changes they made.	algorithm, logical flow, use of appropriate flowchart symbols, and clear, understandable structure. **Assessment Criteria**: Ability to identify and fix logical errors, understanding of algorithm flow, and proper explanation of corrections.	• Compassion • Analytical Skills • Active Participation
Week 6	ILO1, ILO2 [CLO1] [CLO2]	**PRELIMINARY EXAMINATION**	• Examination on the topics discussed during the Prelim Term	• Computer based Lecture Exam	• Skills and Competencies • Critical Thinking • Honesty

Week 7-9	ILO1: Describe the basic concepts of data communication, including how data is transmitted from one device to another through various communication channels.. [CLO 1][CLO2] ILO2: identify and explain the key components of data communication systems, such as the sender, receiver, transmission medium, message, and protocols. [CLO 1][CLO2] ILO3: Understand the role of communication protocols (e.g., TCP/IP, HTTP, FTP) in ensuring reliable and structured data transmission, explaining their purpose and functionality in a network. [CLO 1][CLO2]	**IV. DATA COMMUNICATION AND DATA PROCESSING** • Components of Data Processing and Networking	• Assign small groups of student's different communication protocols (Each group researches their assigned protocol and presents its function, usage, and importance in data communication. • Assign students roles such as sender, receiver, and transmission medium. They simulate the process of data communication (e.g., sending a message through an email or a chat application). They must discuss what happens when there are delays, errors, or incorrect protocols.	**Task**: Each group must submit a written report detailing their findings on the assigned communication protocol, including: • **Introduction**: Background and development of the protocol. • **Function**: Detailed explanation of how the protocol works. • **Usage**: Real-world examples and applications where the protocol is commonly used. • **Importance**: The significance of the protocol in data communication. **Assessment Criteria**: Clarity of explanation, depth of research, use of relevant examples, and presentation of technical details in an understandable way.	• Collaboration • Innovation • Humility
	ILO1: Describe the basic components	**V. SOFTWARE: SYSTEMS AND**	• Students will install different operating	**Assessment Criteria**: Correct configuration	• Active participation

Week 10-11	and functionalities of both system software (e.g., operating systems) and application software (e.g., word processors, web browsers). [CLO 1][CLO2] ILO2: Explain how operating systems manage hardware resources, provide user interfaces, and serve as a platform for running application software.	**APPLICATIONS** • Software ○ Systems Software ○ Applications Software • Operating Systems ○ Functions of an Operating System ○ Structure of an Operating System • Programs that make up an Operating System ○ Control Programs ○ Service Programs • Survey of an Operating Systems ○ MS-DOS ○ The Windows Family of Operating Systems: Brief History of Windows ○ UNIX and Linux • An Overview of Application Software ○ Word Processing ○ Spreadsheets ○ Database Management ○ Graphics Software	systems (e.g., Linux, Windows) on virtual machines (VMware, VirtualBox). They will explore the interfaces, file management systems, and task managers of each operating system. **Task**: Students will write a comparative analysis of different middleware and utility software tools, highlighting their functions, use cases, and advantages. This may include database middleware, web servers, antivirus, or disk cleanup utilities.	of the operating system, file management operations, and system setup. **Assessment Criteria**: Depth of analysis, understanding of software functionalities, and real-world application examples.	and involvement • Active participation and involvement • Critical Thinking • Patience • Understanding • Critical Analysis • Logical Thinking
	LO2, LO3, LO9		• Examination on the	• Computer based	• Skills and

Week 12		**MIDTERM EXAMINATION**	topics discussed during the Midterm	Lecture Exam	Competencies • Critical Thinking • Honesty
Week 13-14	ILO1. Be able to describe the key principles of software development, including problem-solving, design thinking, and the software development life cycle (SDLC). [CLO 2][CLO3] ILO3. Evaluate software solutions based on performance, usability, and security criteria, identifying areas for improvement and optimization.	**VI. SOFTWARE DEVELOPMENT** • Program Planning and Development ○ Defining the Problem ○ Designing a Solution ○ Writing the Program ○ Compiling, Debugging and Testing the Program ○ Documentation of the Program • Structured Design ○ Top-down vs. Bottom-up ○ Structured Design using Flowcharts • Programming Paradigm ○ What is a Paradigm? ○ Major Programming Paradigms • Programming Language ○ Machine Language	• Organize students into teams and simulate an Agile software development cycle. Each team will take on different roles (e.g., Scrum Master, Developer, Tester) and complete tasks in iterations or "sprints."	**Task**: After participating in an Agile simulation, students will write a reflection paper discussing their experience with Agile methodologies, what they learned about teamwork and iterative development, and how Agile differs from traditional models like Waterfall. **Assessment Criteria**: Depth of reflection, understanding of Agile methodologies, ability to connect personal experience to Agile principles.	• Innovation • Compassion • Interiority • Appreciation • Understanding • Accuracy • Critical Thinking • Patience

		○ Low-level Language ○ High-level Programming Language ○ Examples of High-level Language • Interpreters and Compilers ○ Compiler ○ Interpreter			
Week 15	LO1: Identify and explain the functions of key networking devices, including routers, switches, hubs, modems, and access points, and how they facilitate computer interconnections. [CLO 2][CLO3] LO9: Compare wired (e.g., Ethernet) and wireless (e.g., Wi-Fi, Bluetooth) connections in terms of speed, security, and reliability, and evaluate the appropriate contexts for each. [CLO 2][CLO3] LO10: Demonstrate the Filipino-Catholic	**VII. COMPUTER INTERCONNECTIONS** • Basic Principles of Networks ○ Use of Computer Networks ○ Communication Channels • Distributed Systems ○ Distributed Data Processing ○ LAN's and WAN's • The Internet and the World Wide Web ○ Brief History ○ Current Trends and Issues	**Task**: Students will be assessed on their ability to set up a basic computer network (wired or wireless). They must configure devices, assign IP addresses, and ensure successful data transmission. Students will work in groups to compare wired and wireless network connections. Each group will present the pros and cons of each type, focusing on speed, security, and reliability.	**Assessment Criteria**: Correctness of network setup, proper configuration of IP addresses, and successful communication between devices. **Assessment Criteria**: Quality of research, clarity of presentation, and depth of comparison between wired and wireless technologies.	• Leadership • Appreciation • Interiority • Accuracy • Critical Thinking • Patience

	Augustinian Spirituality through value webbing. [CLO4]				
Week 16-17	ILO1. Describe the components of multimedia systems (e.g., text, images, audio, video) and how they are integrated into digital applications. [CLO 2][CLO3] ILO2. Apply design principles in creating multimedia content that incorporates various elements like graphics, animations, sound, and videos, enhancing user experience. [CLO 2][CLO3]	**VIII. MULTIMEDIA AND VIRTUAL REALITY** • Multimedia and Virtual Reality • Applications of Virtual Reality	**Activity**: Students will design and create a basic 3D model or animated object using 3D software such as Blender, Maya, or Unity. They will apply textures, lighting, and movement to their models. **Task**: Students will create a 3D model and present it to the class, explaining the design process, challenges, and how they envision the model being used in a VR environment.	• Group Activity • Recitation **Assessment Criteria**: Technical skills in 3D modeling, creativity, clarity of explanation, and relevance to VR applications.	• Active participation and involvement • Understanding • Courage • Interiority
Week 18	LO4, LO5, LO6, LO7, LO8, LO9	**FINAL EXAMINATION**	• Examination on the topics discussed during the Final Term	• Computer based Machine Problem Exam • Final Project Presentation	• Skills and Competencies • Critical Thinking • Honesty

VIII. COURSE REQUIREMENTS :

1. Class attendance
2. Recitations

3. Quizzes
4. Examinations
5. Compilation of activity reports

IX. SUGGESTED REFERENCES

A. BOOKS

Awari, G. K., and Warjurkar, S. V. (2022). *Ethics in information technology: a practical guide.* CRC Press. TK 5105.878 A93 2022

Bernard, M. (2022). *Future skills: the 20 skills and competencies everyone needs to succeed in a digital world.* John Wiley & Sons, Inc. T 65.3 M37 2022

Delfanti, A. (2019). *Introduction to digital media.* Wiley. QA 76.575 D48 2019

B. INTERNET REFERENCES

MIT Open Courseware (2024). "Introduction to Design Computing" https://ocw.mit.edu/courses/4-500-introduction-to-design-computing-fall-2008/download/ Massachusetts Institute of Technology

Coursera 92024). "Introduction to Computers" https://www.coursera.org/learn/introduction-to-computers

C. JOURNALS

Primiero, G. (2019). Introduction." *On the Foundations of Computing*, pp. 1–4., doi:10.1093/oso/9780198835646.003.0001.

Jianwen, C. and Wakil, K. (2019), "A model for evaluating the vital factors affecting cloud computing adoption: Analysis of the services sector", *Kybernetes*, doi:10.1108/K-06-2019-0434

X. GRADING SYSTEM

1.	Class Standing	60%
2.	Examination	40 %
3.	Term Grade	CS (60%) + TE (40%)
4.	Final Rating	Prelim Grade + Midterm Grade + Final Grade

Prepared by:  **ENGR. GLAZEN CARREY M. PEREZ** Faculty/Date	Checked and Reviewed by: **EDNA F. DAYAO, DIT** Dean-Date **ROSALYN S. GALVEZ, PhD., LPT** Asst. Vice President for Academic Affairs-Date References Certified by: **MARIA MERCEDEZ M. HAZ, PhD, RL** LIC Director/Date	Approved by: **JAY MARK B. SANTOS, PhD, LPT, CSSWB** Vice President for Academic Affairs-Date

Revision Date:	
Modifications: __Learning Outcomes __Course Contents __Suggested Learning Activities __Suggested Assessment __Expected Values to be Integrated __Course Requirements __References __Grading System	